

House Price Prediction — Findings Summary

1. Which features influence price the most?

Square footage (area) is the dominant driver, accounting for roughly 47% of the model's predictive weight. Bathrooms rank second (about 15%) and carry the largest dollar-for-dollar impact in the linear model (11 lakh per extra bathroom). Air conditioning, parking spaces, number of stories, and bedroom count each contribute smaller, roughly similar amounts. Guestrooms and basements have the weakest individual influence.

2. How accurate is the model?

Model	R ² Score	MAE	RMSE
Linear Regression	0.54	1127483	1514173
Random Forest	0.48	1150445	1610415

The linear regression model explains about 54% of the variation in price ($R^2 \approx 0.54$) and slightly outperforms the random forest model. In practical terms, predictions are typically off by roughly 9.7–10 lakh on a property worth 47 lakh on average — about 20% error. That makes the model useful for a rough estimate, but not precise enough to set an exact asking price. The residual error likely reflects missing variables such as location, property age, and condition, which are not present in this dataset.

3. What was surprising?

- Bathrooms outweigh bedrooms as a price driver — unexpected, since bedroom count is usually the first thing buyers ask about. This may reflect buyers reading bathroom count as a signal of a modern, renovated home rather than just extra space.
- Air conditioning is one of the strongest single amenities, outweighing basement presence and main-road access — a sizeable price swing for a simple yes/no feature, likely driven by climate demand rather than a luxury signal.

4. Recommendation for a real estate business

Lead listings with square footage and bathroom count — they matter most to buyers and to price. For improving a property pre-sale, prioritize retrofitting air conditioning over remodeling a basement: AC is a comparatively low-cost upgrade but shows an outsized effect on price in this data, while basement renovation shows a much weaker return. AC upgrades likely offer better ROI per rupee spent than basement work.
